

Scheduling with Time Aware Shaping in Integrated Ethernet and Wi-Fi Networks

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Abstract

Communication technologies play a significant role in systems engineering. The proliferation of connected embedded systems has resulted in Internet of Things applications. Many of these systems are safety or mission critical. For example, in smart manufacturing or health, reliability and low latency are significantly important. A deterministic communication system with real-time capability is essential for ensuring safety in those circumstances. Patient monitoring and remote surgery are two such examples. The maximum permitted delay for video in remote surgery is 200 ms and the minimum allowed latency for tactile feedback is 10 ms. The prior efforts of VLAN tagging over Ethernet are insufficient to ensure determinism and reliability. The upcoming IEEE Time-Sensitive Networking (TSN) standards have enabled determinism by regulating latency, jitter, and packet loss across different types of flows. However, currently, it is only widely available on Ethernet. The paper explores a scheduling approach for an integrated Ethernet and WiFi scenario dealing with mixed-priority traffic that leads to minimal violations of the end-to-end delay requirement. With WiFi TSN bridges and access points equipped with Time Aware Shapers, the proposed scheduling strategy is able to offer determinism while satisfying the requirements of a specific class of IoT applications.

Keywords: Time Sensitive Networking, Ethernet, WiFi, Reinforcement Learning, Scheduling, Simulation.

1 Introduction

Communication technologies play a significant role in systems engineering for various sectors such as Industrial, Smart Health, or Vehicular systems. In the last decade, the proliferation of connected embedded systems led to the Internet of Medical Things (IoMT) in Smart Healthcare or the Industrial Internet of Things (IIoT) in Smart factories. Some of these applications are safety-critical and pose stringent reliability and bounded low-latency communication requirements [1]. The safety-critical nature implies that there is very little tolerance for failures. These systems rely heavily on the communication networks to achieve tolerance. For example, in Industrial Automation systems, Quality of Control (QoC) is dependent on Quality of Service (QoS) provided by the communication network [2].

A deterministic communication system with real-time capability is a prerequisite for safety in Smart healthcare. For example, in a remote surgery scenario, a video lag of a maximum of 200 ms may be allowed, but a much longer delay could have dangerous consequences. In a scenario where vital signs are monitored for patients, 250 ms is the maximum allowed latency [3]. Apart from latency concerns, packet loss above a threshold may result in subpar Quality of Experience (QoE), which may cause incorrect interpretations and lead to wrong judgments.

The existing methods for handling real-time scenarios in communication networks are based on application-specific requirements, such as EtherCAT [4] in industrial automation or Avionics Full Duplex Switched Ethernet (AFDX) in aviation. These technologies pose a challenge to integration with open standards like Ethernet. A seamless integration would mean that the critical flows that require bounded latency share the same communication channel with flows that require lower timing constraints. Having mixed traffic in real-time applications is a challenge today.

The rise of IEEE Time-Sensitive Networking (TSN), a set of standards, is a major step towards ensuring highly reliable and deterministic behavior in networks [5]. TSN focuses on Timing and Synchronization, High Reliability, Bounded low latency, and Resource management. Moreover, the TSN standard allows flows with different requirements and priorities. Although TSN is used in industrial applications and has a reasonable level of maturity over Ethernet, it is yet to be included in wireless standards.

The target of the TSN standards is to ensure zero congestion loss and bounded latency for time-sensitive traffic. The standard applies a combination of per-traffic class scheduling and shaping, resource reservation, pre-emption, and replication to achieve these targets.

TSN defines scheduling and shaping aspects as part of IEEE 802.1Qbv - Enhancements for Scheduled Traffic (TAS) and 802.1Qcr Asynchronous Traffic Shaping(ATS). Traffic shaping relies on queuing, management of queue and transmission selection. Time-schedule-based, credit-based and asynchronous are different approaches

available for traffic shaping. Resource reservation is defined in 802.1Qcc while reservation aspects like Stream Reservation Protocol and Frame Preemption are defined in 802.1Qbu. Frame Replication and Elimination as in IEEE 802.1CB provides Reliability whereas IEEE 802.1Q part defines the bridge functionality. A frame stands for an instance of a message on a particular network link, while a stream stands for information distribution across sender and receiver. Since our focus is on meeting the timing requirements of requests from an end-to-end perspective, handling traffic as streams is preferred in real-time scenarios.

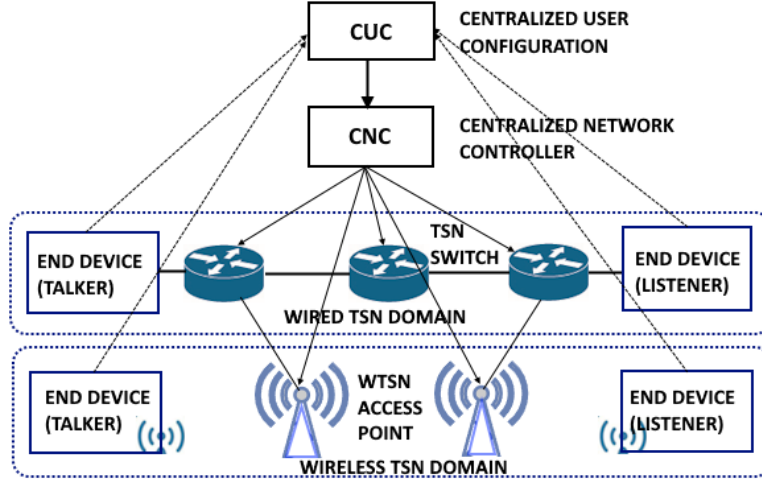


Fig. 1: TSN Layout and Configuration

In a time-aware system, time synchronisation plays an important role. Time synchronisation over Precision Time Protocol (IEEE 1588) is part of the standard in 802.1QAS-Rev. Though initial TSN standards targeted Ethernet (IEEE 802.1) being built on Virtual LAN (VLAN) bridging (IEEE 802.1Q) [1], TSN is considered key for future wireless standards like WiFi7 for ensuring QoS.

TSN devices use stream encoding and Virtual LAN(VLAN) tagging to create prioritized traffic streams. As defined in IEEE 802.1 Qbv, every switch in a TSN Network has a set of priority queues with gates that permit packet transmission when they are open. A Gate Control List (GCL) defined for the switch controls the opening and closing. A user or administrator in the TSN process defines the workflows, their time criticality, and priority considerations, as shown in Fig. 1.

The Centralized User Configuration (CUC) gathers the specifications. The Centralized Network Controller (CNC) collects the user configuration from CUC. The configuration, along with the topology, serves as inputs to CNC for preparing a flow schedule. The computed flow schedule comprises the priority of the flow, its start time, the duration, and, in the event of a Time Aware Shaper (TAS), the precise timeslots at which gates would open and close. The computed flow schedule is used by CNC to

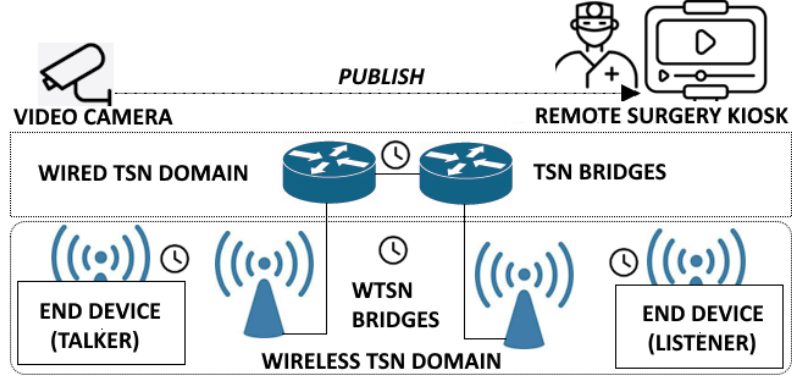


Fig. 2: Health care scenario: Publishing from Video Camera to Display

configure the GCL on each TSN bridge. The IEEE 802.1Qci Per-Stream Filtering and Policing (PSFP) standard, enables filtering and decision-making per stream. It offers QoS protection to streams allowing non-conformance detection and taking mitigation actions.

Fig.2 illustrates a typical scenario in remote surgery where the network includes both Ethernet and wireless end devices. The scheduling strategy needs to consider the uncertainties of wireless. Along with hard real-time systems, the strategy must handle a mixed traffic scenario where delay-tolerant systems like monitoring and analytics and soft real-time systems may co-exist in the network. Among wireless protocols, WiFi is the first to be considered for inclusion in time-sensitive standards.

Future wireless standards like WiFi 7 consider supporting TSN as a crucial strategy for guaranteeing QoS. There are multiple research challenges in providing TSN functionalities in a hybrid network of wired and wireless endpoints, which include;

- How to support Time-aware scheduling in Wireless access points by generating network configurations similar to those supported in Ethernet?
- How do deterministic solutions cater to mixed traffic scenarios in a hybrid network with wired and wireless devices?

From a design point of view, modeling and simulation of systems are key to a well-designed system. This is even more significant in the upcoming Cyber-Physical systems. In the context of Industrial Networks or Hospital Networks, the ability to predict how a communication network will behave is key to system design and development. Communication networks and simulation models for these systems enable significant capabilities to know emergent behavior beforehand so that we can avoid failure. Our work builds on the work done by others in Time Sensitive Networks and OMNET++ for simulation modeling of integrated Ethernet and WiFi with time-sensitive network characteristics.

The real-time application requirements are often categorized into Class A, Class B and Class C [6]. Class A takes into consideration applications like high-quality audio video, soft real-time control, and autonomous guided vehicles. The latency needs of

Class A range from 10 to 50 ms. Class B applications include Augmented Reality (AR)/Virtual Reality (VR), gaming, Human Machine Interfaces (HMI), Smart Health and hard real-time control. The latency needs of Class B is between 1 to 10 ms. Class C application examples include Hard real-time control and motion control in industrial IoT. The permitted latency in these systems is 1 ms or less. Most of them come with high reliability needs.

This paper proposes a reinforcement-based scheduling strategy for an integrated network with wired and wireless TSN endpoints and makes the following contributions:

- It identifies key capabilities to be built on WiFi to enable TSN. The work builds these capabilities in a simulation environment where Time-Aware Shapers (TAS) are extended to wireless access points.
- It demonstrates the notion of scheduling request streams with different end-to-end delay requirements and priorities.
- Our experimental results demonstrate that the proposed scheduling strategy can achieve an End-to-End delay of less than 50 ms and a sub-millisecond jitter for mixed traffic scenarios satisfying Class A traffic.

The paper is organized as follows: Section 2 analyzes related work in the area of deterministic communications. Our main methodology is explained in Section 3 and simulation results are presented in Section 4. The paper is concluded in Section 5.

2 Related Work

The related work is analyzed from the perspectives of schedule generation and wireless TSN solutions. A schedule meeting the resource, timeliness, and precedence requirements is considered realistic or feasible. Most modern techniques target minimizing schedule makespan while meeting the requirements.

Several methods have been used for time aware schedule generation which includes constraint solvers using Satisfiability Modulo Theories (SMT) . The solvers check for collisions across flows. The approach is static and does not scale. Yinzhi Lu et al. address intelligent ultra-low latency communication in their work. [7]. Based on the combinability of flows, the authors propose dynamic scheduling for non-time-sensitive flows and deterministic scheduling based on collision avoidance. In their study, Zhiwei Feng et al. address scheduling in the context of lowering packet loss [8]. The authors suggest a proactive fault-tolerant method, a drawback being the waste of network resources.

Heuristics and Metaheuristic approaches are discussed in [9] and [10] respectively. Heuristics is found to performing better than constraint-based methods. A technique for supporting no-wait time aware scheduling with Size Step Scheduling for Packet (PASS) is presented by Junhong Min et al. as part of metaheuristic approach. The authors compare the performance of evolutionary techniques. They observe that Artificial Ecosystem-based Optimization produces a lower cost among these methods.

In their paper [11], Maryam et al. use Genetic Algorithms (GA) for population creation and fitness function validation. The combinatorial technique fares well and shows a significant improvement in schedulability.

In real-time schedule generation, Chilukuri et al. provide RECCE, a deep REinforcement learning technique for joint routing and scheduling that takes wireless and centralized control into consideration [13]. Their work focuses on finding the best timeslots for TDMA-based Medium Access Control in order to fulfill deadlines. When a viable schedule is unachievable, the strategy attempts to guarantee that the least number of packets miss the deadline.

A key element of a TSN solution is Traffic Shaping for enabling time awareness in WiFi. Authors in [14] present solutions for wireless TSN using OMNeT++ or NeSTiNg Simulation Models. The strategy validates scheduling in the existing IEEE 802.11ac protocol; however, does not deal with latency bound at the access points. A strategy for IEEE 802.1Qbv scheduling of wireless time-critical industrial traffic in a collaborative robotic scenario is proposed in [15], which efficiently utilizes the channel bandwidth. This work focuses on WiFi with limited integration with wired. Seijo et al. [16] address challenges in the integration of wired and WTSN (Wireless TSN) and propose the W-SHARP architecture. Their work considers the wireless stack with 802.15.4 and takes a Linux-based approach using FPGA. They have demonstrated a 3 TSN hops delay of 150 microseconds. However, this is a hardware-driven approach. Another work [17] using Matlab-based simulation of TSN and IEEE 802.1, demonstrates low latency real-time communications among collaborative robots. In an attempt to ensure timeliness over the air, Cavalcanti et al, demonstrate feasibility of Wireless systems like IEEE 802.11ax and 5G for real-time industrial communication using Intel Arria FPGA platform. While the paper set requirements for classes A, B and C, it does not consider joint scheduling of wired and wireless [18]. Channel access which is key for enabling low latency in WiFi scenarios is explored by Stefano Avallone et al [19]. A comparison with the important related work is summarized in Table 1. By analyzing all these works, we identify these as major gaps:

1. A schedule generation strategy integrating Ethernet and WiFi is missing.
2. End-to-End Latency Analysis over a Wired-WiFi integrated system with multiple WiFi nodes and domain translation is not addressed.
3. The support for integrated Wired-WiFi in simulation is missing.

To the best of our knowledge, our work is one of the first of its kind, integrating Ethernet and WiFi with multiple domain translations.

3 Method

In this section, the strategy for enabling Time awareness over WiFi and access point is described. Then, we cover in detail the scenarios handled in our experiments. Enabling Time awareness over WiFi is a pre-requisite for integrating wired and wireless in simulation. The solution is implemented in a simulation environment using OMNET++. OMNET++ is, a Discrete Event Simulator library and framework created with C++ and Qt framework libraries. A more detailed description of the approach is provided in the next subsections.

Ref	Focus	Nw. Model	Performance	Hardware or Simulation	Wired	Wi-Fi	Gaps
[7]	Theoretical Approach	Many switches and sub-scribers	-	Hardware	✓	×	Ethernet only, Static scheduling
[8]	TSN Scheduling with Fault Tolerance	Transient Fault model	-	Hardware	✓	×	Ethernet only scenario, Static scheduling
[9]	TSN Schedule generation	Maximum 9 sub-scribers	-	Simulation	✓	×	Ethernet only Static scheduling
[10]	TSN Schedule generation	NSF network topology	Metaheuristic with the best performance gain	Simulation	×	✓	Static scheduling, Ethernet only
[11]	TSN Schedule generation	Mesh and ring topology	31% Makespan improvement with Genetic algorithm	Hardware	✓	×	Static scheduling in Ethernet only
[12]	TSN Schedule generation	4 switches and 5 sub-scribers	-	Simulation	✓	×	Ethernet only Static scheduling
[13]	Realtime Scheduling	Max 50 nodes	-	Simulation	✓	×	WiFi only scenario TDMA used
[14]	Scheduling with TAS	Many Wired and one WiFi node	Identifies WiFi Access Point as a bottleneck	Simulation	✓	✓	At Access point no TAS support
[15]	Time Aware Shaping WiFi	Ethernet plus WiFi topology	5 ms delay with 99% confidence interval	Hardware	✓	✓	No multiple domain (Wired-WiFi) translation
[16]	Time Aware WiFi	Ethernet plus WiFi topology	150 μs	Hardware	✓	✓	Only Hardware Solution, not demonstrated in larger networks
This work	Time Aware WiFi	Ethernet plus WiFi topology	< 50 ms delay	Simulation	✓	✓	Integrating Wired and WiFi with multiple domain translation (WiFi and Wired)

Table 1: Related work comparison

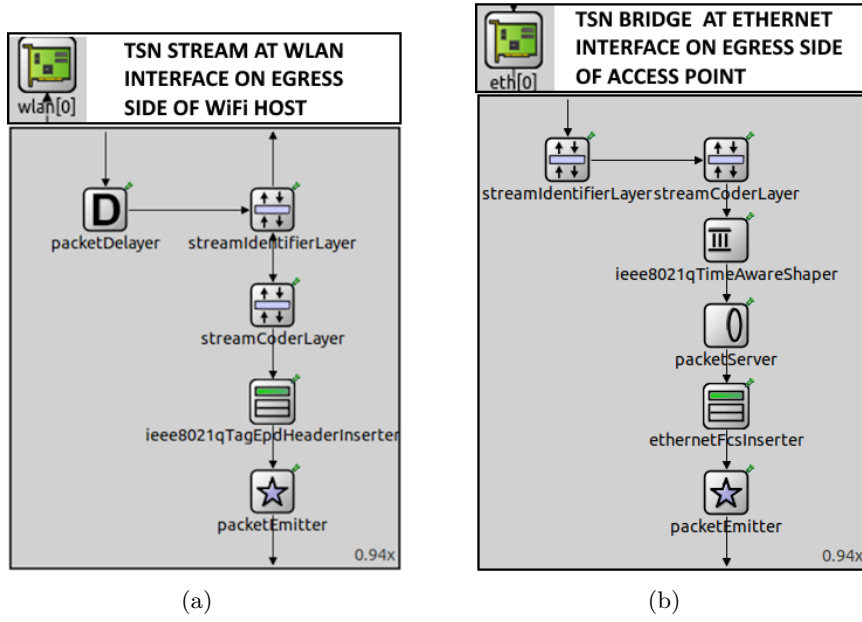


Fig. 3: (a) TSN layer at Source in WiFi (b) TSN Layer at Bridge or Access Point in WiFi

We identify the following three main steps involved in enabling TSN capabilities across an Ethernet-WiFi hybrid network:

- Organizing flows into streams
- Bridge capability to translate domains
- Incorporate Time Aware Shaping to enable time sensitivity

3.1 Enabling TSN Characteristics in Wireless Endpoints

To ensure priority and traffic isolation in mixed-traffic scenarios, the flows are required to be organized as streams. Unlike flows, network streams are considered end-to-end with a source and destination pair and mapped to priority class and type.

Organizing flows into streams:

These are encoded at the *source* where the streams get initiated. To enable streams, we added a *tsnLayerSource* to the source WLAN interface in the INET library of OMNET++. The *tsnLayerSource* combines capabilities of stream identification, encoding and adding IEEE 802.1Q tag to the packet to ensure end-to-end handling. The modifications are made to support TSN capabilities at a source WiFi host, as illustrated in Fig. 3(a).

Bridge Capability for Domain Translation:

From a Time-sensitive network perspective, Ethernet and WiFi are considered two different domains as they exhibit different characteristics. Whenever the streams

cross domains, i.e., from Ethernet to WiFi and vice versa, they undergo a translation to reflect the native domain characteristics, e.g., IEEE 802.1Q tag, Frame Check Sequence, and timing information. This has to be done at Access Points which then communicate with the Switches and vice versa. In this case, it is done by integrating functionality in WLAN layers and LAN interface layers. The addition of a *tsnLayer-Bridge* to the source WLAN interface represents an important step towards the hybrid network. The modifications to support TSN at Access Point, otherwise termed as Bridge, are presented in detail in Fig. 3(b).

Time Aware Shaping:

The AccessPoints need to be enhanced with Time Aware Shaping functionalities to handle time-sensitive stream scheduling. This is done in the INET layer by integrating Time Aware Shaping as a *tsnLayer* in WLAN interfaces.

3.2 Scenarios and Experimentation

Each of the experiments performed comprises of the following steps:

1. Identify Scenarios for experimentation
2. Simulate the network flow scenarios
3. Time Aware Schedule Generation
4. Simulation Results Check
5. Provide Feedback to Reinforcement Learning in order to update states and rewards

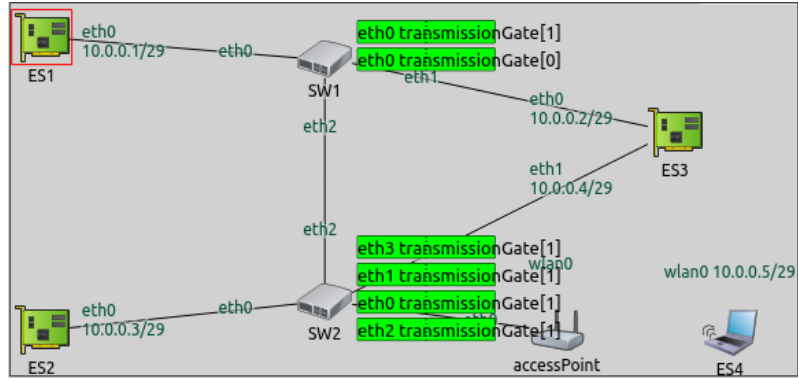
We consider two scenarios to validate the proposed solution. The initial simulation is without time-aware schedules. Then time-aware schedules are prepared, and the simulation is performed. The results are analyzed, and the feedback are used to improve the learning. The simulation scenarios use the following setup. The simulation machine has 12 Gen Intel i5 2.5 GHz CPU with 16 GB Memory. The Wired Links are 100 Mbps bandwidth while Wireless links are 54 Mbps. The video stream of size 500B is having high priority (4) and Best Effort (BE) stream of size 1000B is mapped to lower priority. All flows are Unicast and generated using UDP packets.

These two scenarios are based on IEEE 802.11 Real-Time Applications TIG Report and the Use Cases IEC/IEEE 60802 for Industrial Automation.

3.2.1 Scenario 1: Wired endpoint communicates with Wireless Endpoint

We consider a scenario in Industrial Automation where the control can publish the temperature and pressure values to HMI. Here, the HMI is a wireless endpoint inside a control room or operator room. Vitals reporting in Smart Health, is a similar scenario. Scenario 1 traffic model involves two flows of video and the best effort type from ES1 to ES3 and ES2 to ES3. Another video flow is sent from ES2 to the access point. The wireless host ES3 acts as a receiver for video transmission from the ES2 node. The network topology is depicted in Fig.4. The simulation parameters are :

- Cycle duration: 1s
- Release times: Best Effort 500 ms, Video 250 ms
- Block Acknowledgement: No



- Transmit Opportunity (TXOP) for reserving channel: No

$$\{ES1 \rightarrow SW1 \rightarrow ES3\}$$

Similarly, the traffic from ES2 to ES3 is supposed to follow the path:

$$\{ES2 \rightarrow SW2 \rightarrow ES3\}$$

The third flow is initiated from ES4 which is a WiFi Host and it follows the path:

$$\{ES4 \rightarrow SW2 \rightarrow accessPoint \rightarrow ES2\}$$

3.2.2 Scenario 2: Presence of Multiple Wireless Endpoints

In this experiment, we consider multiple cross-overs from wired to wireless domains. In a Remote surgery case, the videos are reported over the wireless to a room where doctors may control the surgery. Feedback control in industrial automation can also follow a similar case.

The scenario is illustrated earlier in the introduction as Fig. 2. It considers a network topology where WiFi-based TSN devices (talkers) at one end talk to WTSN (Wireless TSN) Bridge (Access Point), which is then connected to an Ethernet TSN Bridge (Switch). This Ethernet TSN bridge is later connected to a WTSN Bridge which is talking to a TSN End Device over wireless. Thus, the experiment involves multiple translations from the Wireless TSN Domain to the Ethernet-based Wired TSN Domain.

The corresponding network topology in the simulation is depicted in Fig. 5. It has two TSN Talkers - `tsnWiFiHost1` and `tsnWifiHost2` and other end points are the two TSN Listeners - `tsnWiFiHost3` and `tsnWiFiHost4`. TSN Talkers are both associated to `tsnAccessPoint1` and both of the TSN Listeners are associated to `tsnAccessPoint2`. The two access points are connected by TSN switches - `tsnSwth1` and `tsnSwitch2`. The TSN talkers support stream identification and encoding, whereas `tsnAccessPoint1`

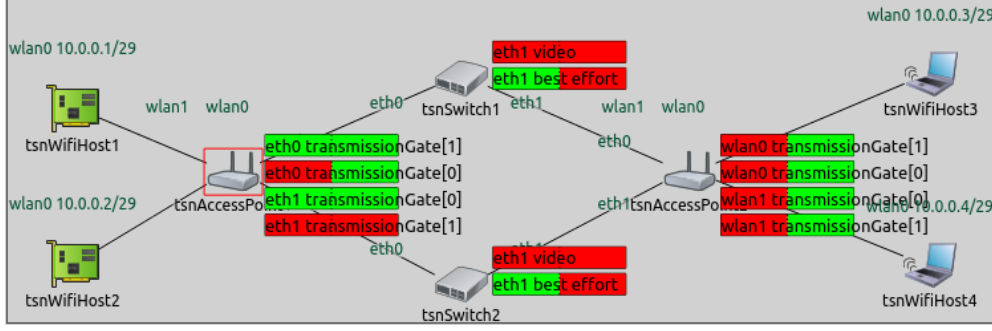


Fig. 5: Scenario 2 Topology

supports bridging and adding QTag when it translates to and from Ethernet. The simulation parameters are :

- Cycle duration: 100 ms
- Release times: Best Effort 10 ms, Video 20 ms
- Block Acknowledgement: Yes
- Transmit Opportunity (TXOP) for reserving channel: Yes

The traffic flows in this scenario include two high-priority video flows and two Best Effort (BE) flows. The traffic from tsnWiFiHost1 to tsnWiFiHost3 follows the path:

$$\{tsnWiFiHost1 \rightarrow tsnAccessPoint1 \rightarrow tsnSwitch1 \rightarrow tsnAccessPoint2 \rightarrow tsnWiFiHost3\}$$

Similarly, the traffic from tsnWiFiHost2 to tsnWiFiHost4 is to follow the path:

$$\{tsnWiFiHost2 \rightarrow tsnAccessPoint1 \rightarrow tsnSwitch1 \rightarrow tsnAccessPoint2 \rightarrow tsnWiFiHost4\}$$

It is noted that Class A requirements need to be met for both scenarios.

3.3 Time-Aware Schedule Generation

We need to consider a global schedule across the topology that can provide timeslots in which the gates of the time-aware shapers should open and close. Deep Reinforcement Learning (DRL) is used to arrive at this schedule. The Network model and the traffic model are the inputs towards the DRL algorithm. The Markov Decision Process (MDP) formulation for the DRL algorithm considers the state space with nodes having a feature vector consisting of minimum time to completion, minimum time to completion at a hop, minimum hops to completion, and flows as action space. The reward design includes high negative rewards for schedules that miss deadlines. The DRL algorithm builds on our previous work [20].

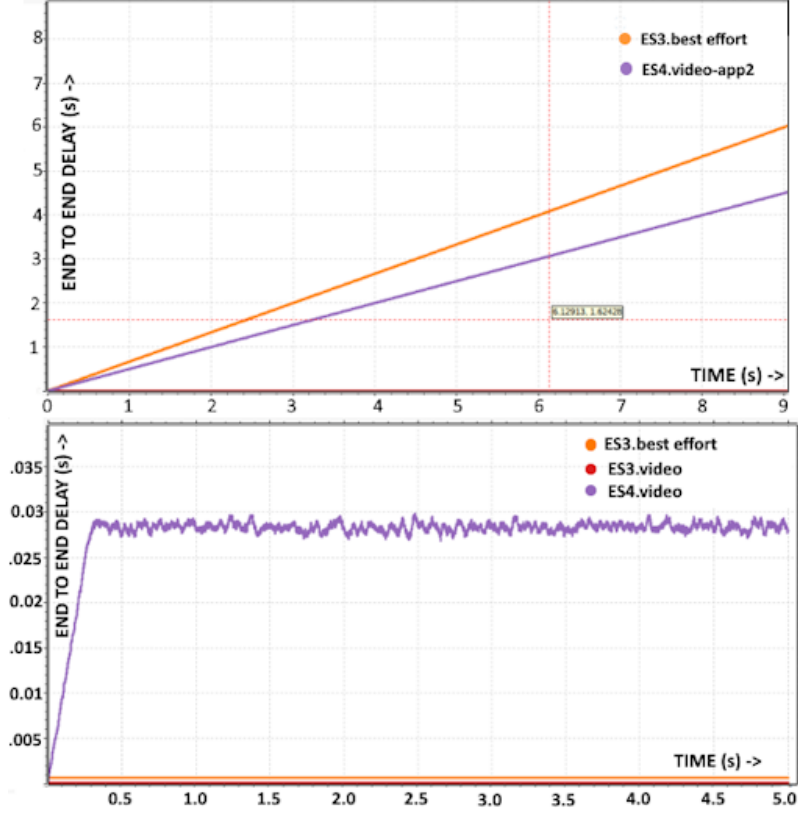


Fig. 6: Scenario 1: Without and With including Wireless in TSN stream

4 Results and Discussions

The main statistic observed in the simulation is the Max Bit Lifetime Per Packet which represents the End-to-End delay in OMNET++. The results are analyzed from the perspectives of two metrics - 1) how well the schedule adheres to deadlines and 2) jitter of delivered packets.

We compare our work with *always open gate* scheduling where packets are sent on a first come first serve (FCFS) basis and the gates at TSN switches are always open. For Scenario 1, the results with always open gates scheduling are illustrated in Fig. 6. The figure illustrates that the delay for video over wireless (ES4.video-app2.sink) and best-effort traffic increases linearly with time. After including wireless TSN capabilities at the access point, the results from simulation show delay and jitter falling in bounds. In Scenario 1, the delay is as low as 30 ms. Given that the flows have deadlines of 200 ms and 300 ms, feasible schedules are obtained from the Deep Reinforcement Learning algorithm.

Next, the results are analyzed for Scenario 2 without any schedule where the switching gates are always open. Fig. 7 shows the end-to-end delay and jitter for this

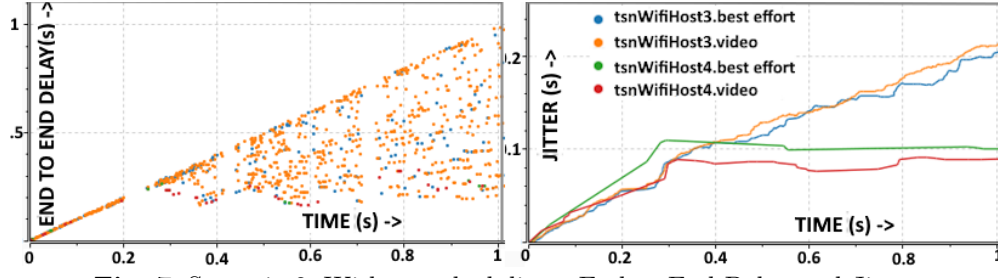


Fig. 7: Scenario 2: Without scheduling - End to End Delay and Jitter

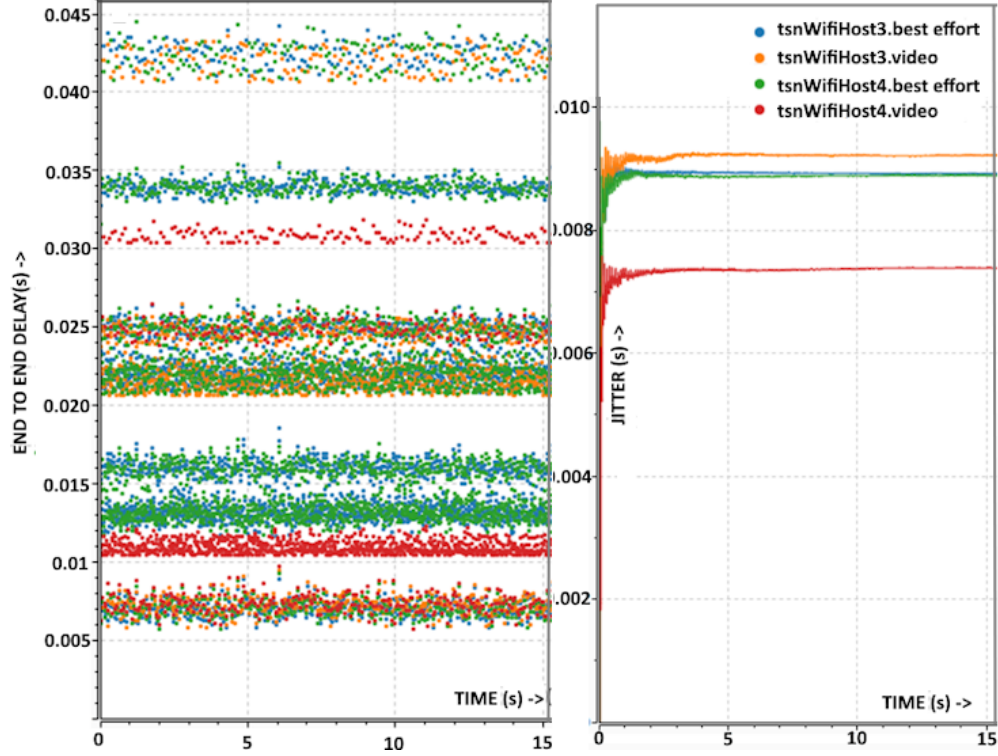


Fig. 8: With TSN scheduling - End to End Delay and Jitter

scenario. Fig. 8 illustrates the End to End delays and jitter after applying the gate scheduling using our algorithm. It can be observed that in spite of multiple domain translations, the delay is less than 45 ms.

The feedback to the Deep Reinforcement Learning Algorithm is in terms of Rewards, where deadline misses lead to a big negative reward, and meeting deadlines leads to big positive reward.

Key observation:

The experimental results of the integration of Ethernet-WiFi with a time-aware schedule demonstrate that we are able to bring the end-to-end delay to less than 50 ms which is the requirement of Class A. This ensures that with TSN scheduling and a Deep Reinforcement Learning-based algorithm, we could satisfy many soft real-time applications in Industrial and IoMT scenarios.

5 Conclusion and Future Work

In this paper, we have identified the key enablers for TSN on WiFi and addressed the challenges in the wireless medium. Our work extends the simulation environment to support integrated wired and wireless scenarios. Through our scheduling approach and simulations, we could demonstrate that Class A requirements can be met in an integrated wired-wireless scenario with mixed traffic. This enables many soft real-time applications in Industrial and IoMT scenarios. In the next steps, we plan to build dynamic capabilities in scheduling, enable WiFi controlled channel access capabilities, and bring down latency to cater to the real-time requirements of Class B and Class C.

For the academia, the addition of time awareness on WiFi in simulator open up new avenues of experimentation. The next steps identified also set clear directions of research. The industry could use simulator for co-design to build newer products with WiFi integration and use the principles to take advantage of scheduling to optimize their products. With reinforcement learning, the system can be enable cyber-physical approach and improve continuously.

Declarations

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- Conflict of Interest: On behalf of all authors, the corresponding author states that there is no conflict of interest.

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